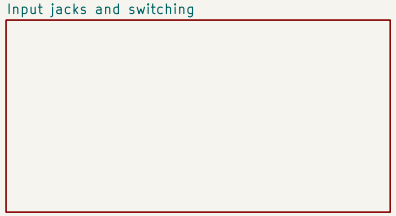
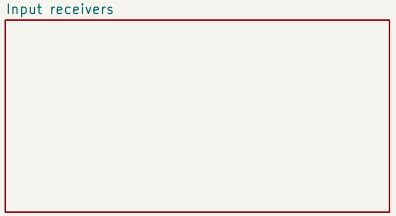
Plug & Roll main board. Each block of the board is one sub-sheet.

Input jacks and switching



File: input-jacks.kicad_sch

Input receivers



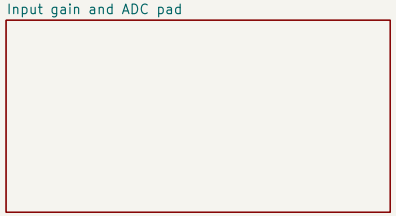
File: input.kicad_sch

INST buffer



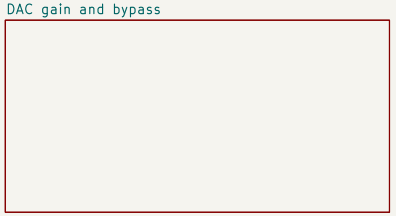
File: input-inst.kicad_sch

Input gain and ADC pad



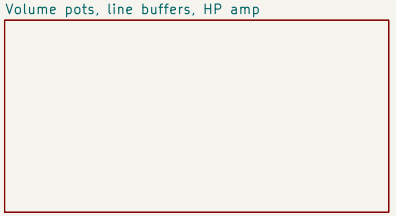
File: input-gain.kicad_sch

DAC gain and bypass



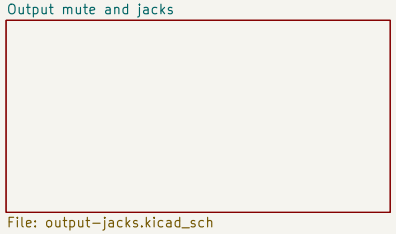
File: output-dac.kicad_sch

Volume pots, line buffers, HP amp



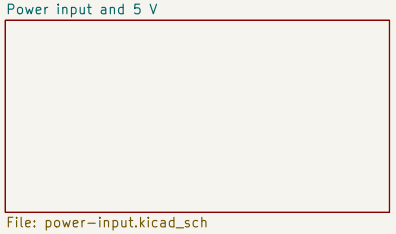
File: output-pots.kicad_sch

Output mute and jacks



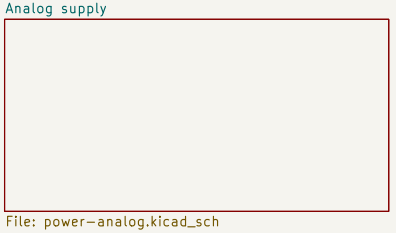
File: output-jacks.kicad_sch

Power input and 5 V



File: power-input.kicad_sch

Analog supply



File: power-analog.kicad_sch

Sheet: /
File: main-board.kicad_sch

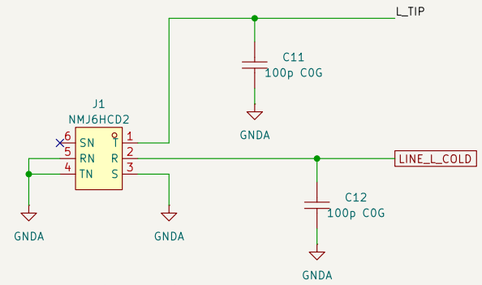
Title: Plug & Roll – Main board

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/10

INPUT JACKS

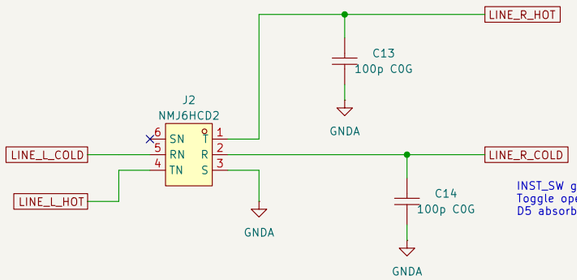
Tip = hot (+), ring = cold (−), sleeve = analog ground.
TN / RN / SN are switch contacts that rest on T / R / S while the jack is empty.

LEFT input



Empty jack: tip and ring are grounded (silent input).

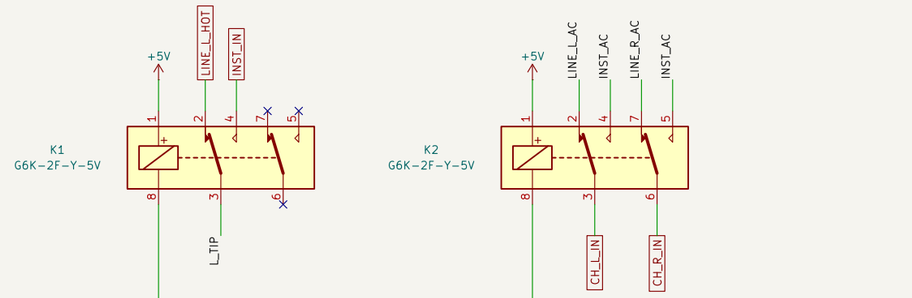
RIGHT input



Empty jack: the Right receiver gets the Left signal (mono).
Purely analog, the Daisy is not involved.

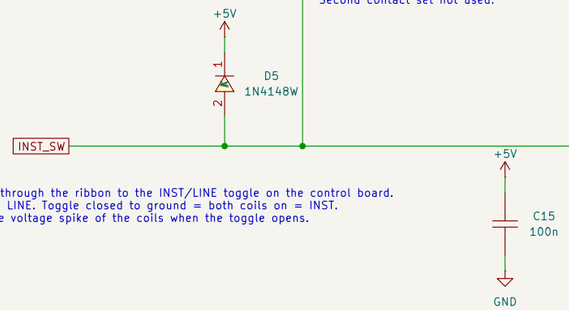
INST / LINE SWITCHING

Relays drawn released (no coil current) = LINE. Panel toggle closed = coils on = INST.



K1: Left tip to the LINE receiver or to the INST buffer.
Second contact set not used.

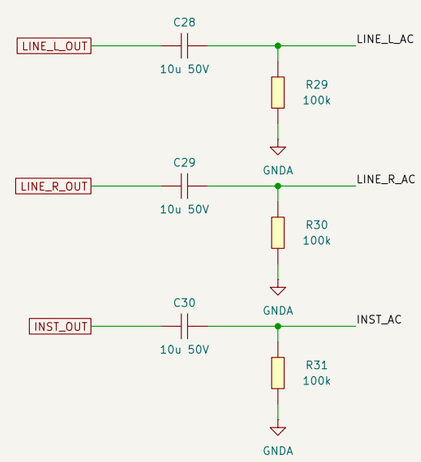
K2: each channel takes its LINE receiver or the INST buffer.



INST_SW goes through the ribbon to the INST/LINE toggle on the control board.
Toggle open = LINE. Toggle closed to ground = both coils on = INST.
D5 absorbs the voltage spike of the coils when the toggle opens.

5 V decoupling, placed at the relays

Coupling capacitors in front of K2



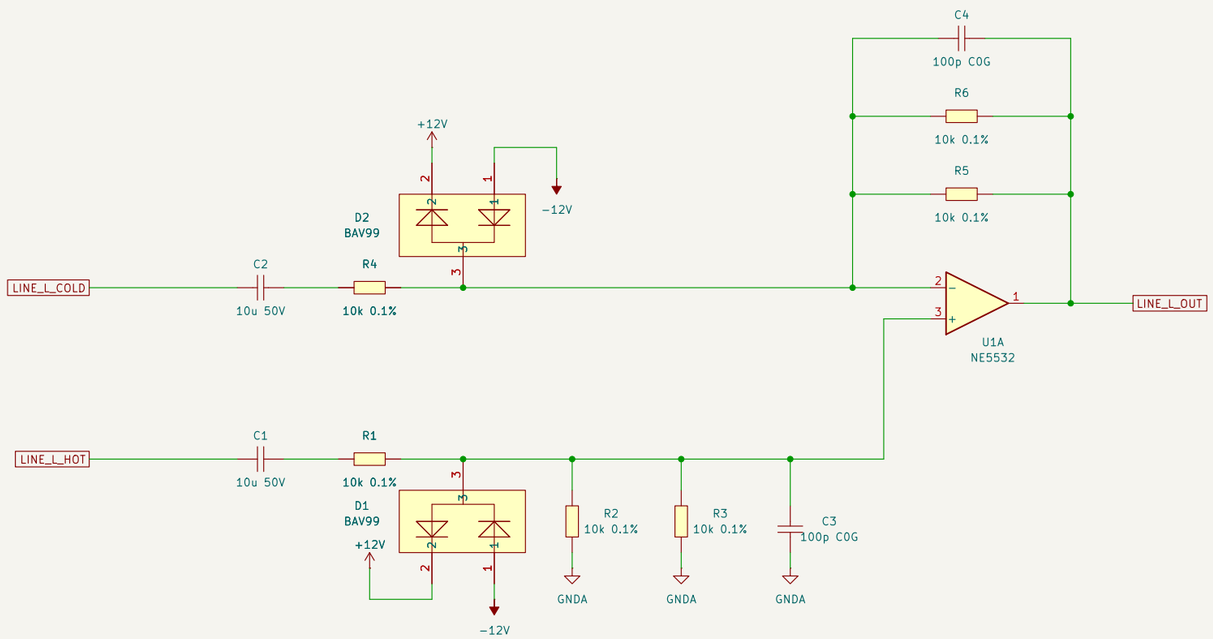
The three sources reach K2 without their DC offset,
so switching INST / LINE gives no DC step (click).

Sheet: /Input jacks and switching/
File: input−jacks.kicad_sch

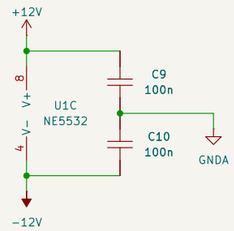
Title: Plug & Roll − Main board − Input jacks and INST/LINE switching

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/10

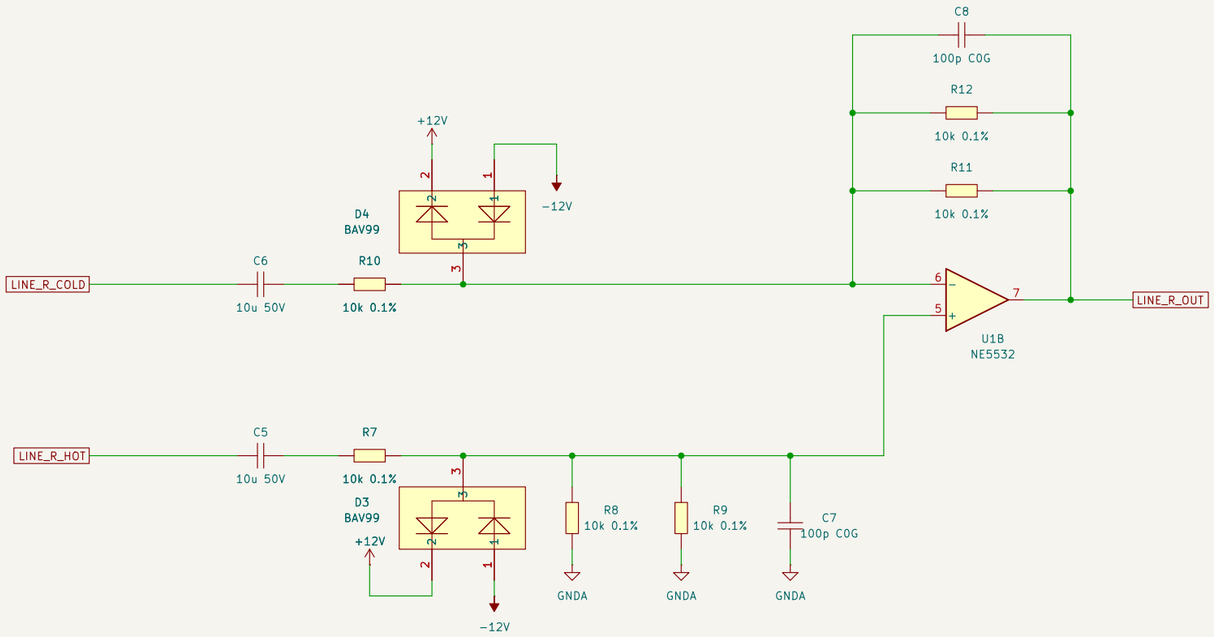
LINE receiver L: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



U1 supply and decoupling



LINE receiver R: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



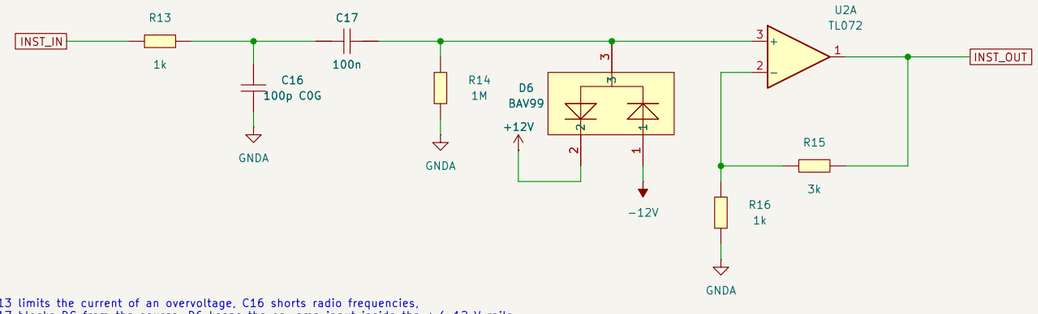
Sheet: /Input receivers/
File: input.kicad_sch

Title: Plug & Roll – Main board – Input receivers

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/10

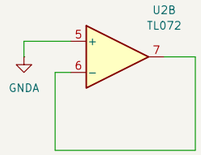
INST BUFFER

Instrument (guitar) input, used only in INST mode. Input resistance 1 Mohm, gain x4 (+12 dB) = 1 + R15/R16.

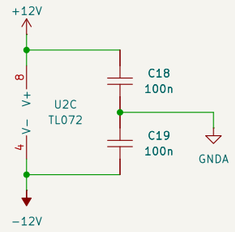


R13 limits the current of an overvoltage, C16 shorts radio frequencies,
C17 blocks DC from the source, D6 keeps the op-amp input inside the +/−12 V rails.

Unused half of U2



U2 supply and decoupling



Sheet: /INST buffer/
File: input-inst.kicad_sch

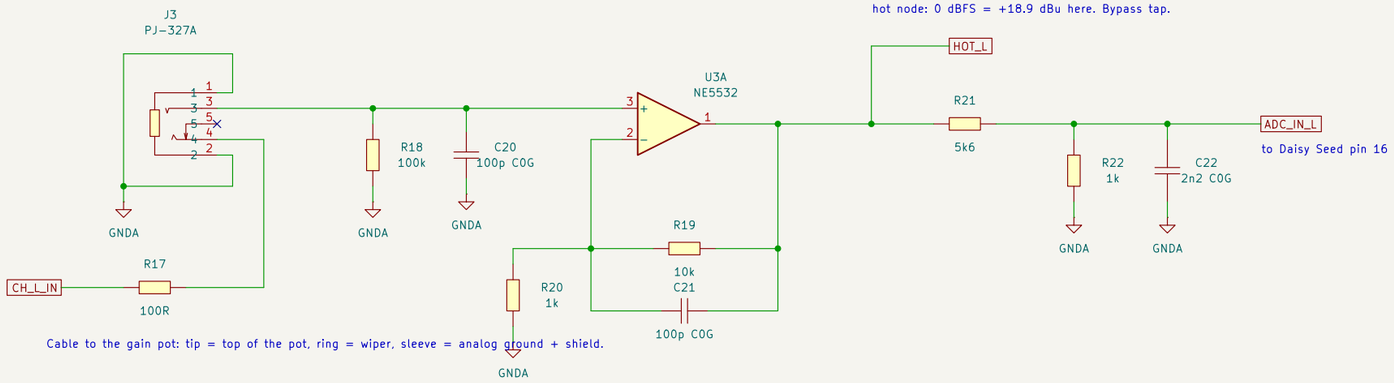
Title: Plug & Roll – Main board – INST buffer

Size: A3 | Date: | **Rev: 0.1**
KiCad E.D.A. 10.0.6 | Id: 4/10

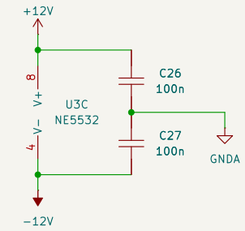
INPUT GAIN AND ADC PAD

The gain pot (10k dual, audio taper) is on the control board. One shielded 3.5 mm stereo cable per channel connects it to J3 / J4.
Pad 5k6 / 1k brings the hot node down to codec level and limits voltage and current at the Daisy pin. No bias: the Seed has its own input capacitor.

LEFT channel: gain pot, x11 (+20.8 dB), pad to the Daisy



U3 supply and decoupling



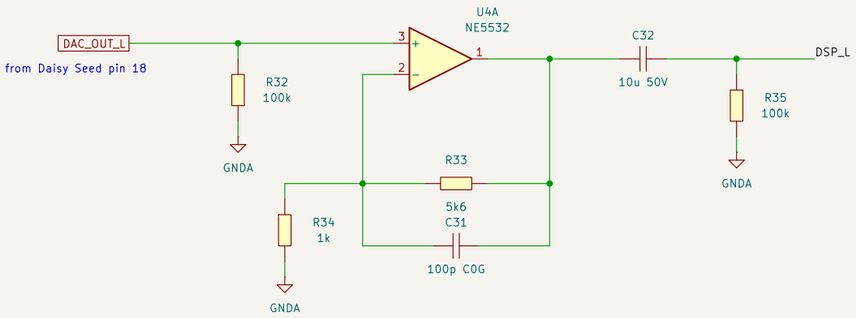
RIGHT channel: gain pot, x11 (+20.8 dB), pad to the Daisy

Sheet: /Input gain and ADC pad/		
File: input-gain.kicad_sch		
Title: Plug & Roll — Main board — Input gain and ADC pad		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/10

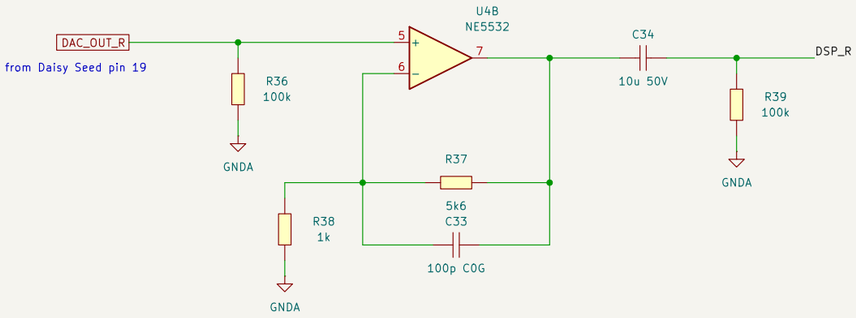
DAC GAIN AND GLOBAL BYPASS

The Daisy output (1.0 Vrms at 0 dBFS) is amplified to the level of the hot node, so the effect path and the analog bypass path reach the bypass relay at the same level. No bias: the Seed has its own output capacitor.

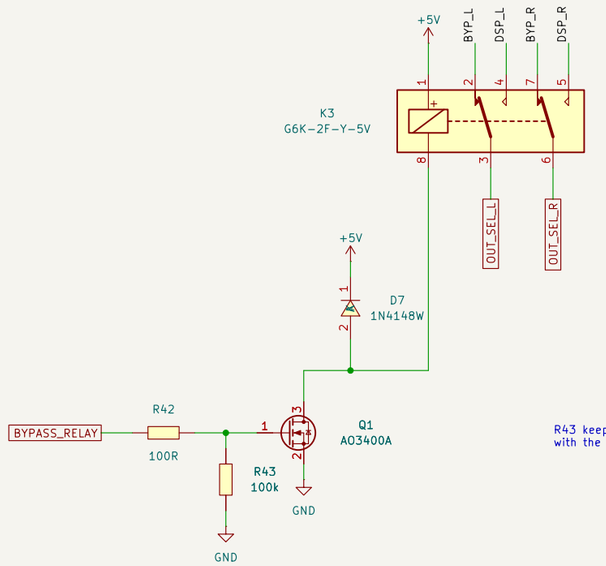
LEFT channel: DAC gain x6.6 (+16.4 dB) = 1 + 5k6 / 1k



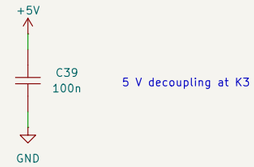
RIGHT channel: DAC gain x6.6 (+16.4 dB) = 1 + 5k6 / 1k



BYPASS RELAY. Drawn released (no coil current) = analog bypass. BYPASS_RELAY high = effect path.

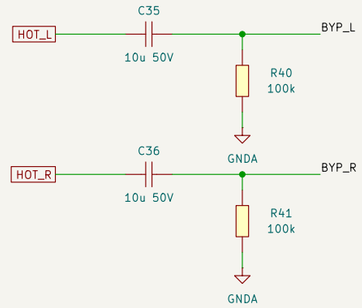


R43 keeps the transistor off until the Daisy drives the line: at power-on, in reset and with the Seed removed the relay stays released = bypass. D7 absorbs the coil's switch-off spike.

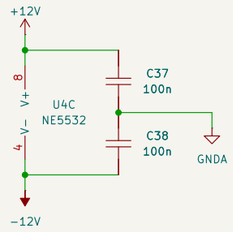


5 V decoupling at K3

Analog bypass signal from the input section



U4 supply and decoupling



Sheet: /DAC gain and bypass/
File: output-dac.kicad_sch

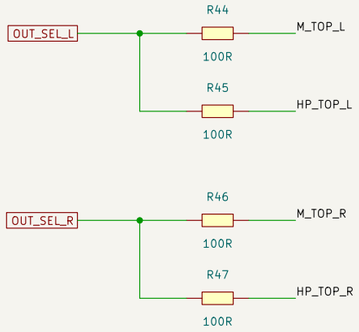
Title: Plug & Roll – Main board – DAC gain and global bypass

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/10

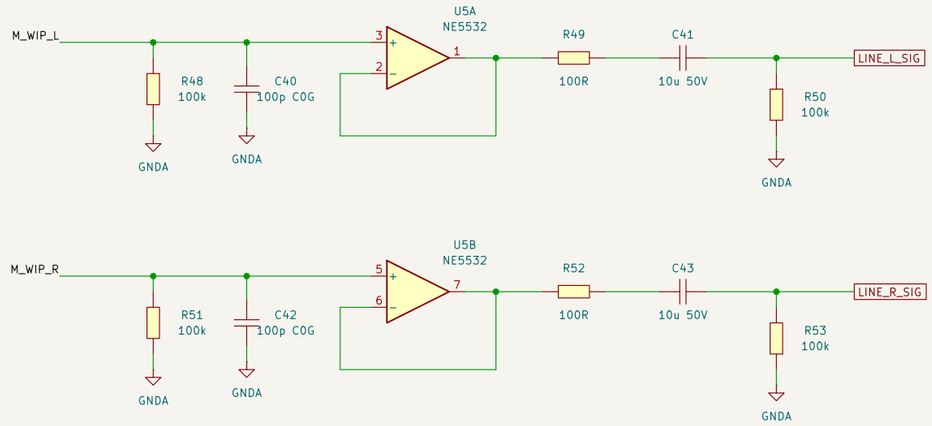
VOLUME POTS, LINE BUFFERS, HEADPHONE AMPLIFIER

Master and HP volume pots (10k dual, audio taper) are on the control board. One 16-wire ribbon connects them to J5:
odd pins = signals, even pins = analog ground, so every signal wire has a ground wire next to it.

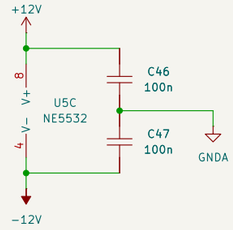
From the bypass relay to the tops of the pots



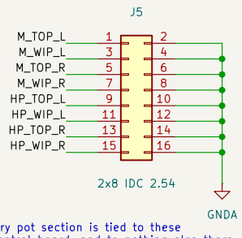
LINE BUFFERS (unity gain) after the master volume



U5 supply and decoupling

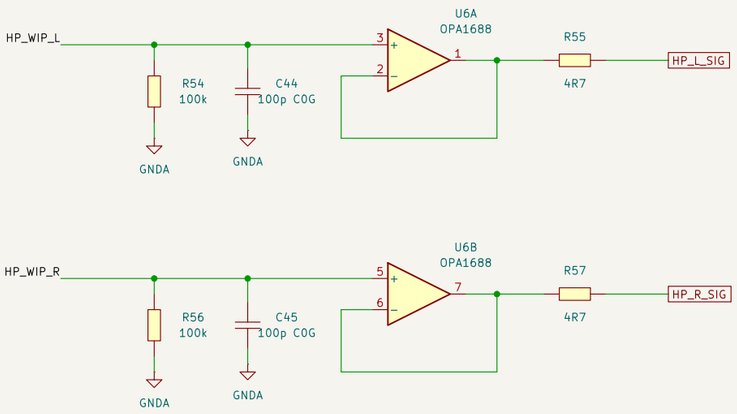


J5: ribbon to the master and HP pots



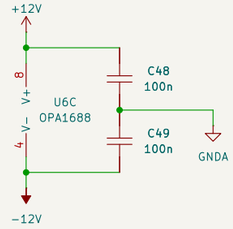
The bottom end of every pot section is tied to these
ground wires on the control board, and to nothing else there.

HEADPHONE AMPLIFIER (OPA1688, unity gain) after the HP volume



4R7 in series: protects the amplifier while a plug is pushed in (it shorts the contacts on its way)
and keeps the cable capacitance away from the output. Output impedance about 5 ohm.

U6 supply and decoupling

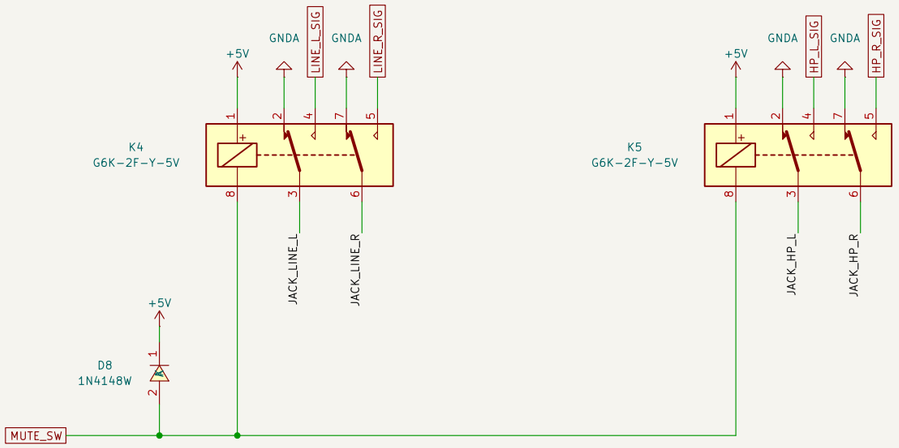


Sheet: /Volume pots, line buffers, HP amp/		
File: output-pots.kicad_sch		
Title: Plug & Roll — Main board — Volume pots, line buffers, headphone amp		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 7/10

OUTPUT MUTE RELAYS AND JACKS

Relays drawn released (no coil current) = muted: every output contact of a jack is tied to ground, the amplifiers are disconnected.
MUTE_SW pulled to ground by the power-on delay circuit (power section) = coils on = outputs connected.

K4: line outputs

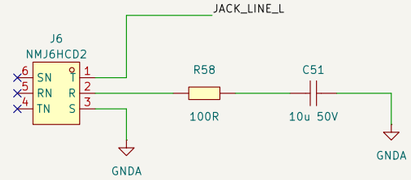


K5: headphones

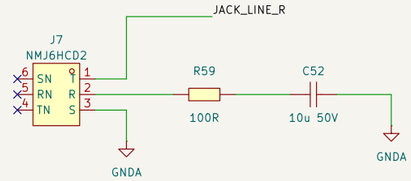


5 V decoupling at K4 / K5

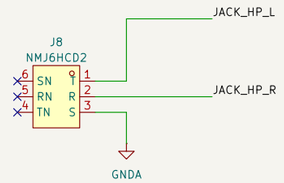
LINE OUT LEFT (impedance balanced)



LINE OUT RIGHT (impedance balanced)



HEADPHONES (tip = left, ring = right)

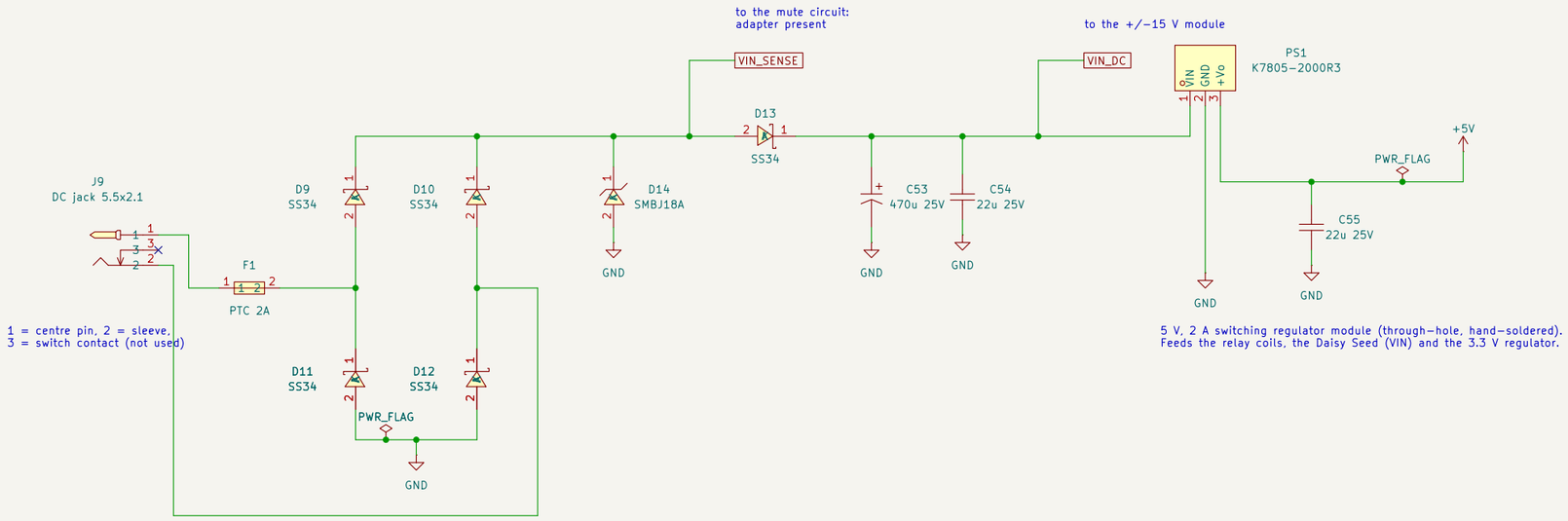


Line outputs: the ring carries no signal. It has the same 100R + 10u to ground as the tip has to the amplifier, so a balanced input at the other end rejects hum picked up on the cable.

Sheet: /Output mute and jacks/		
File: output-jacks.kicad_sch		
Title: Plug & Roll – Main board – Output mute relays and jacks		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 8/10

POWER INPUT AND 5 V

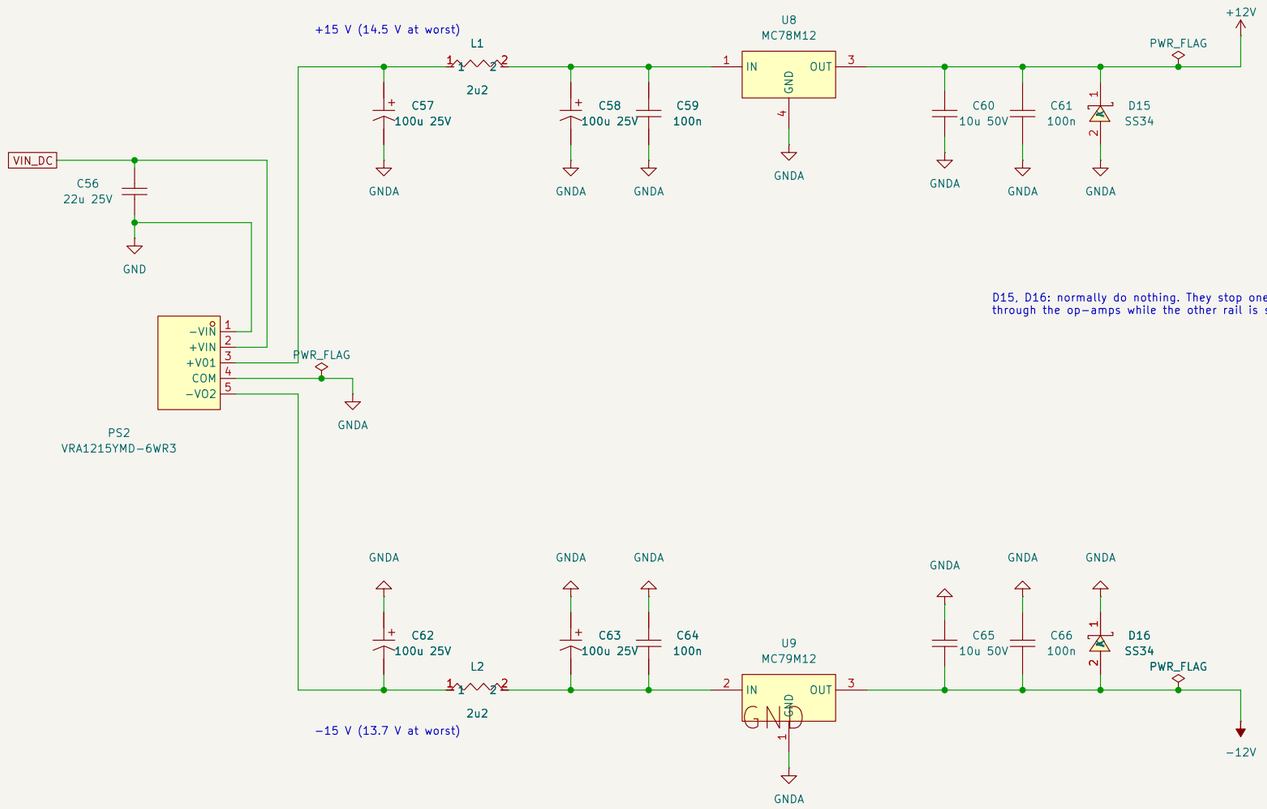
Adapter: 12 V DC, 2 A or more, regulated, plug 5.5 x 2.1 mm, either polarity.
F1 limits a fault current. The four diodes D9–D12 pass the current the right way round whatever the polarity of the plug.
D14 clamps voltage spikes (plugging in a live adapter). D13 lets VIN_SENSE fall at once when the plug is pulled, while C53 keeps the converters fed.



Sheet: /Power input and 5 V/		
File: power−input.kicad_sch		
Title: Plug & Roll — Main board — Power input and 5 V		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 9/10

ANALOG SUPPLY: +/−15 V MODULE, FILTERS, +/−12 V REGULATORS

PS2 is an isolated converter: its output side has its own ground, which becomes the analog ground GNDA of the board.
Each output passes a filter (100 uF, 2.2 uH, 100 uF) and a fixed regulator. The op-amp rails must stay inside 11.4 … 12.6 V.
At most 220 uF may hang on each module output (YLPTEC datasheet): 2 x 100 uF here.



D15, D16: normally do nothing. They stop one rail being pulled past ground through the op-amps while the other rail is still rising at power-up.

Electrolytics on the negative rail have their + terminal on GNDA.

Sheet: /Analog supply/	
File: power-analog.kicad_sch	
Title: Plug & Roll — Main board — Analog supply +/−12 V	
Size: A3	Date:
KiCad E.D.A. 10.0.6	**Rev: 0.1**
	Id: 10/10